Understanding the Mechanisms of Fast Hyperparameter Transfer

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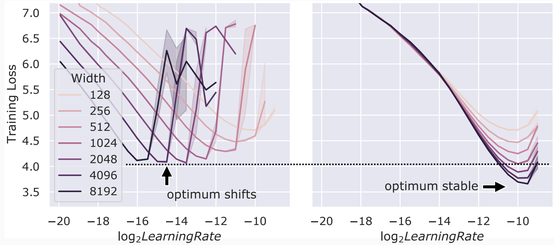
Introduction to μ-Transfer

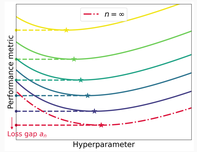
Hyperparameter tuning is **brutally expensive** at scale. The **Maximal Update Parameterization (μP)** offers a cheap alternative: tune on a small proxy model, then *transfer* the HPs to a large run. *But when does this actually work — and why?*



Standard parameterization (left): the optimal HP shifts with width. μP (right): the optimum is stable — so small-model tuning transfers.

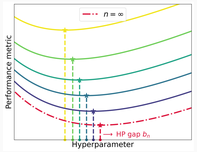
Fast transfer

What does it mean for transfer to "work"? μP guarantees optimal HPs have a width-independent limit but says *nothing about rates*. At width n we track three gaps — and one geometric fact:



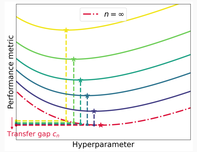
Loss gap a_n

finite vs. infinite width



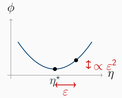
HP gap b_n

how far apart the best HPs are



Transfer cost c_n

loss paid for using the wrong HP



Baseline

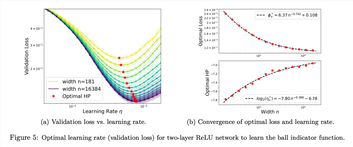
$b_n = \mathcal{O}(\sqrt{a_n})$ near a smooth min.

Fast transfer: b_n shrinks *strictly faster* than $\sqrt{a_n}$ — beating the baseline. We prove this is *exactly* when transfer is **useful** (asymptotically more compute-efficient than direct tuning). And the baseline is not automatic: a two-layer ReLU under μP matches it *exactly*, so transfer only breaks even there.

Useful-transfer theorem. Let $a_n \sim n^{-\alpha}$, $b_n \sim n^{-\beta}$ near a smooth minimum. Then transferring HPs from width n to ∞ is *asymptotically cheaper* than tuning directly at width n iff

$$\beta > \alpha/2, \qquad \text{equivalently} \qquad b_n = o(\sqrt{a_n}).$$

Why: near a smooth minimum the transfer cost scales as $c_n \propto b_n^2$, while direct tuning pays $\sim a_n$. So transfer wins exactly when $b_n^2 = o(a_n)$ — *the HP gap must shrink faster than* $\sqrt{\text{loss gap}}$.



(a) Validation loss vs. learning rate. (b) Convergence of optimal loss and learning rate.

Figure 5: Optimal learning rate (validation loss) for two-layer ReLU network to learn the ball indicator function.

Two-layer ReLU + μP: $b_n \propto n^{-0.39} \approx \sqrt{a_n}$ — baseline exactly achieved, so fast transfer is not universal under μP. It must come from additional structure.

The mechanism: low-rank trajectory decomposition

Intuition: optimal HPs should depend on network statistics that stabilize with width *faster* than the loss itself. We decompose the stepwise linearized loss change via the eigenvalues of a gradient/update alignment matrix:

Matrix parameter $\boldsymbol{W} \in \mathbb{R}^{m\times n}$, gradient $\boldsymbol{G} \in \mathbb{R}^{m\times n}$, and update $\delta\boldsymbol{W} \in \mathbb{R}^{m\times n}$.

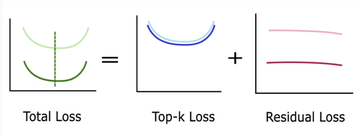
$$\boxed{\begin{aligned} \boldsymbol{A} &:= \tfrac{1}{2}\left(\boldsymbol{G}^\top \delta\boldsymbol{W} + \delta\boldsymbol{W}^\top \boldsymbol{G}\right) && \text{(alignment matrix)} \\ |\lambda_1| &\geq \cdots \geq |\lambda_n| && (\lambda_i \text{ eigenvalues of } \boldsymbol{A}) \end{aligned}}$$

Stepwise linearized loss change for parameter $\boldsymbol{W}$:

$$\delta\phi(\boldsymbol{W}) := \langle \boldsymbol{G}, \delta\boldsymbol{W}\rangle = \mathrm{Tr}(\boldsymbol{A}) = \underbrace{\sum_{i=1}^{k} \lambda_i}_{\delta\phi^k(\boldsymbol{W})} + \underbrace{\sum_{i=k+1}^{n} \lambda_i}_{\delta\phi^{>k}(\boldsymbol{W})}.$$

Summing over layers ℓ and time steps t:

$$\underbrace{\mathcal{L}(\mathbf{w}_T) - \mathcal{L}(\mathbf{w}_0)}_{\text{total loss}} \approx \underbrace{\sum_{\ell=1}^{L}\sum_{t=0}^{T-1} \delta\phi^k\left(\boldsymbol{W}_t^{(\ell)}\right)}_{\text{top-}k\text{ loss}} + \underbrace{\sum_{\ell=1}^{L}\sum_{t=0}^{T-1} \delta\phi^{>k}\left(\boldsymbol{W}_t^{(\ell)}\right)}_{\text{residual loss}}.$$



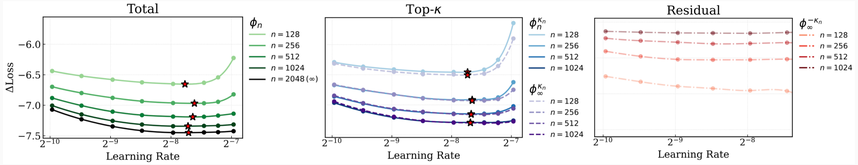
Cartoon: total loss = (U-shaped, width-stable) top-k + (flat) residual.

Conjecture (qualitative). For some k: **(1)** top-k loss is locally strongly convex at its optimum; **(2)** *top-k invariance* — $\phi_n^k \approx \phi_\infty^k$, $\nu_n^{k*} \approx \nu_\infty^{k*}$; **(3)** *residual flatness* — $\nu_n^{k*} \approx \nu_n^*$.

Experiments

We measure the decomposition on an actual LLM training run. The same three-way split shown in the cartoon appears directly in the data: top-k curves overlap across widths and share the total's optimum, while residuals are flat in the LR.

Setup: 4-layer LLaMA · WikiText-103 · Adam · warmup–stable–decay LR · widths 128–2048 ($n = 2048$ as ∞-proxy).



Total ϕ_n

U-shaped; optima shift with width.

Top-k $\phi_n^{k_n}$

Overlap across widths; *same* optimum as total.

Residual $\phi_n^{-k_n}$

Flat in LR; can't move the optimum.

All three conjecture pieces (LSC, invariance, flatness) **verified** on a real LLM run.

In the paper: provable fast transfer for RF ridge regression; transfer behavior of Muon & Dion; a data-centric viewpoint of the top-k loss; a μP failure case. arXiv:2512.22768.

Takeaway. μP alone only guarantees HP transfer — not *useful* transfer. What makes tuning a small proxy worthwhile is a *low-rank structure in training*: a small, width-stable top-k component carries the HP signal, while the rest is flat.